

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI - K K BIRLA GOA CAMPUS

Semester-I (2025/2026)

Analog & Digital VLSI Design (EEE/INSTR F313)

Mid-semester Examination

Date : 09.10.2025

Duration : 90 Minutes

Maximum Marks : 60

Name of the student: Palak Khetrapal ID. No. 2023A8PS0914G

Instructions :

1. Please write final answers for each question in the table provided.
2. Answers without legitimate units will be marked invalid.
3. Overwritten answers will not be rechecked under any circumstances.
4. There are total 5 questions in the paper.

Recheck Request

Question No.	Student's request for rechecking with remarks	Examiner's remarks
1		
2		
3		
4		
5		

Question	Marks	Score
1	13	13
2	14	NA
3	12	10
4	12	00
5	9	00
Total:	60	23

Question 1[13]

Suppose that in Figure 1, $I_1 = 100 \mu\text{A}$, $I_2 = 0.5 \text{ mA}$, and $(W/L)_{1,2,3} = 100/0.5$. Assuming that I_1 and I_2 are implemented with PMOS devices having $(W/L)_p = 50/0.5$,

(a) Calculate the gate bias voltages of M_2 and M_3 (DC voltage at point X and P).

(b) Determine the maximum allowable output voltage swing.

(c) Calculate the overall voltage gain.

Given: $V_{DD} = 3 \text{ V}$, $\mu_n C_{ox} = 135 \mu\text{A/V}^2$, $\mu_p C_{ox} = 40 \mu\text{A/V}^2$, $V_{th,n} = 0.7 \text{ V}$, $|V_{th,p}| = 0.8 \text{ V}$, $\lambda_n = 0.1 \text{ V}^{-1}$ at $L = 0.5 \mu\text{m}$, $\lambda_p = 0.2 \text{ V}^{-1}$ at $L = 0.5 \mu\text{m}$.

[13]

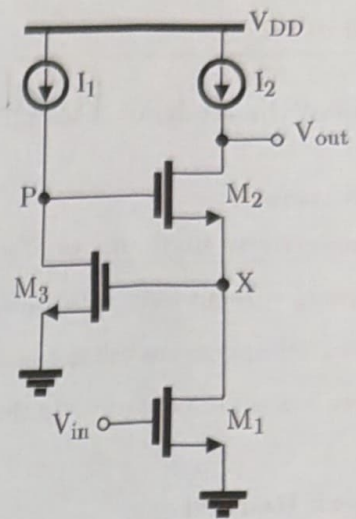


Figure 1: Circuit diagram for Question 1

V_X	0.786V
V_P	1.678V
Maximum allowable output voltage swing	1.5216V
Overall voltage gain	51.726V/V

Important Steps:

$$I_{D3} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_3 (V_{OV3})^2$$

$$V_{OV3} = 0.086$$

$$V_G = 0.786$$

$$I_{D2} = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L}\right)_2 (V_{OV2})^2$$

$$V_{OV2} = 0.0124$$

$$V_{OV2} = V_{GS} - V_t$$

$$V_P - V_{OC} - V_t = V_{OV2}$$

$$V_P = V_{OV2} + V_t + V_{OV2}$$

$$V_G = 1.678$$

$$\begin{aligned} V_{out,max} &= V_{DD} - V_{OV2} \\ &= 3 - 0.5 \\ &= 2.5 \end{aligned}$$

$$\begin{aligned} V_{out,min} &= V_{OV3} + V_{OV2} \\ &= 0.9784 \end{aligned}$$

$$r_{O1} = r_{O2} =$$

$$\begin{aligned} A_v &= -g_{m1} r_{O1} + r_{O2} \\ &= -g_{m1} (g_{m1} r_{O1} r_{O2} + 1 / I_{D2}) \end{aligned}$$

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Question 2[14]

Design (find size of all MOSFETs, value of I and C_c) Miller compensated two stage operational amplifier shown in Figure 2 for following specification:

- DC Gain = 60 dB
- Gain Bandwidth Product=50 MHz
- Phase margin $\geq 60^\circ$
- slew rate = 50 V/ μ s
- ICMR(+) = 1.6 V
- ICMR(-) = 0.9 V
- $C_L = 5$ pF

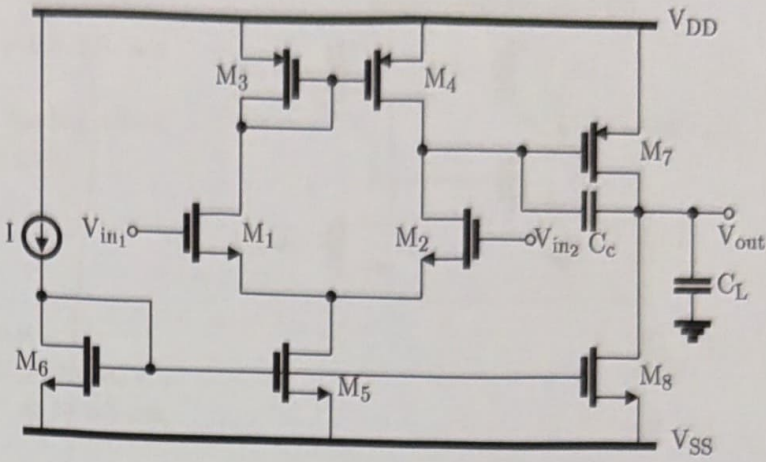


Figure 2: Circuit diagram for Question 2

Use minimum value of compensation capacitor required for 60° phase margin. Assume that current mirror, made up of M_3 and M_4 , is perfect in operation to size M_7 . also M_6 and M_5 are of same size.

Use $\mu_n C_{ox}=300 \mu A/V^2$, $V_{th1}(\max)=0.59$ V, $V_{th1}(\min)=0.47$ V, $\mu_p C_{ox}=60 \mu A/V^2$, $V_{th3}(\max)=-0.51$ V, $V_{DD}=1.8$ V, $V_{SS}=0$ V.

$\left(\frac{W}{L}\right)_{1,2}$	
$\left(\frac{W}{L}\right)_{3,4}$	
$\left(\frac{W}{L}\right)_{5,6}$	
$\left(\frac{W}{L}\right)_7$	
$\left(\frac{W}{L}\right)_8$	
I	
C_c	

Important Steps:

Question 3

[12]

Consider the circuit of Figure 3, where $(W/L)_{1-4}=50/0.5$, $|I_{D1-4}|=0.5$ mA, $R_1=0.8$ k Ω and $R_2=3$ k Ω .

(a) Find the open-loop gain including loading effect.
You can neglect r_{o1} while calculating gain.

(b) Find the feedback factor.

(c) Calculate the closed-loop gain.

(d) Find the closed-loop output impedance.

Given: $V_{DD}=3$ V, $\mu_n C_{ox}=135$ μ A/V², $\mu_p C_{ox}=40$ μ A/V², $V_{th,n}=0.7$ V, $|V_{th,p}|=0.8$ V, $\lambda_n=0.1$ V⁻¹ at $L=0.5$ μ m, $\lambda_p=0.2$ V⁻¹ at $L=0.5$ μ m.

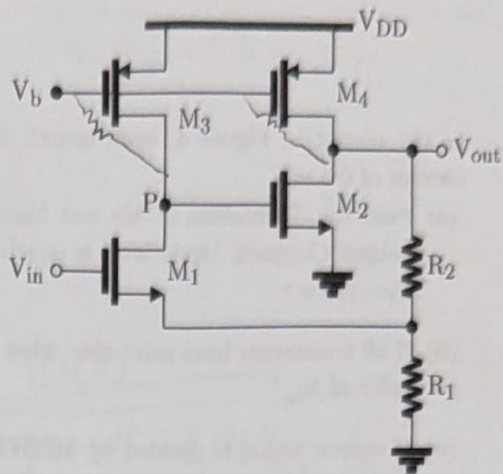


Figure 3: Circuit diagram for Question 3

Open-loop gain	-111.95 V/V
Feedback factor	-0.211 V/V
Closed-loop gain	4.527 V/V
Closed-loop output impedance	111.837 Ω

$$\beta = \frac{V_o}{V_i} = \frac{R_1}{R_1 + R_2} = \frac{0.8}{3.8} = 0.211 \text{ V/V}$$

Important Steps:

~~$$A = g_{m1}(R_1 || R_2 || 1/g_{m3}) \times g_{m2}(R_1 + R_2 || 1/g_{m4})$$~~

$$A_2 = g_{m2}(r_{o4} || r_{o2} || (R_1 + R_2))$$

~~$$A_1 = g_{m1}$$~~

$$A_1 = \frac{-g_{m1}(r_{o3} || r_{o1})}{1 + g_{m1}(R_1 || R_2)}$$

~~$$A_2 = \frac{g_{m2}(r_{o4} || r_{o2})}{1 + g_{m2}(R_1 + R_2)}$$~~

$$g_{m1} = \frac{2I_D}{V_{OV}} = 3.67 \times 10^{-3}$$

$$g_{m2} = g_{m1} = 3.67 \times 10^{-3}$$

$$r_{o3} = r_{o4} = \frac{1}{\lambda_p I_D} = 10 \text{ k}\Omega$$

$$R_1 = 0.8 \text{ k}\Omega$$

$$R_2 = 3 \text{ k}\Omega$$

$$r_{o2} = 20 \text{ k}\Omega$$

$$A = A_1 \times A_2$$

$$= -11.065 \times 10.117$$

$$= -111.95$$

$$A_B = \frac{A}{1 + A\beta}$$

$$R_O = r_{o2} || r_{o4} || (R_1 + R_2) / (1 + \beta A)$$

Question 4 [12]

In the circuit of Figure 4, each branch is biased at a current of 0.5 mA.

- Find the dimensions of M_7 and M_8 such that the output Common Mode level is equal to 1.5 V and $V_P=100$ mV.
- If all transistors have same size, what is the typical value of V_{b1} ?
- If output swing is decided by MOSFETs shown in diagram, what is the maximum differential output voltage swing?

Given: $V_{DD}=3$ V, $\mu_n C_{ox}=135 \mu\text{A}/\text{V}^2$, $\mu_p C_{ox}=40 \mu\text{A}/\text{V}^2$, $V_{th,n}=0.7$ V, $|V_{th,p}|=0.8$ V, $\lambda_n=0.1 \text{ V}^{-1}$ at $L=0.5 \mu\text{m}$, $\lambda_p=0.2 \text{ V}^{-1}$ at $L=0.5 \mu\text{m}$.

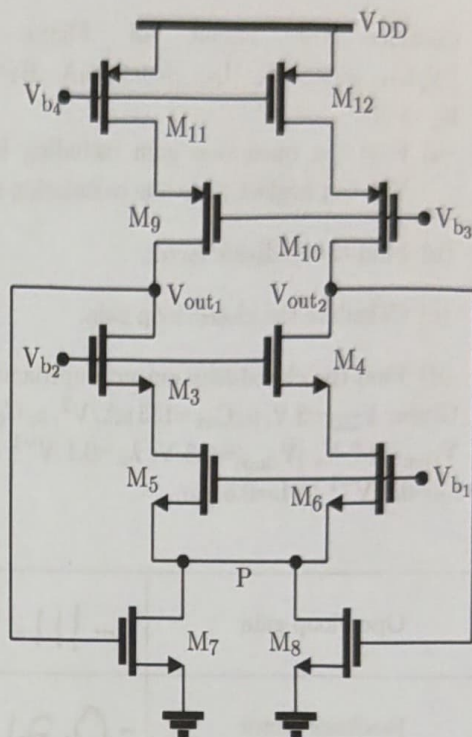


Figure 4: Circuit diagram for Question 4

$\left(\frac{W}{L}\right)_{7,8}$	2962.96	0
Typical value of V_{b1}	0.15V	0
Maximum differential output voltage swing		0

Important Steps:

$$V_G = \frac{1.5}{2} = 0.75$$

$$V_{GS} - V_t = 0.05 = V_{OV}$$

$$V_{OV} = 0.05$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{OV})^2$$

Question 5 [9]

- (a) An amplifier has a forward DC gain of $A_M=1000$ and two poles at ω_{p1} and ω_{p2} . For $\omega_{p1}=1$ MHz, calculate the phase margin of a unity-gain feedback loop if (i) $\omega_{p2}=2\omega_{p1}$ and (ii) $\omega_{p2}=4\omega_{p1}$.

Phase margin for $\omega_{p2}=2\omega_{p1}$	71.565°
Phase margin for $\omega_{p2}=4\omega_{p1}$	58.513°

Important Steps:

$$A_M = 1000$$

$$A_B = \frac{A_M / \left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right)}{1 + \frac{BA}{\left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right)}}$$

$$= \frac{A_M \times \cancel{(1000)}}{s^2 + (\omega_{p1} + \omega_{p2})s + (1 + BA)\omega_{p1}\omega_{p2}}$$

$$\left| \frac{A_M}{1 + j\omega \left(\frac{1}{\omega_{p1}} + \frac{1}{\omega_{p2}}\right) + \frac{\omega^2}{\omega_{p1}\omega_{p2}} + BA} \right| = 1$$

$$\frac{1000}{\sqrt{\left(1 + BA - \frac{\omega^2}{\omega_{p1}\omega_{p2}}\right)^2 + \left(\omega^2 \left(\frac{1}{\omega_{p1}} + \frac{1}{\omega_{p2}}\right)\right)^2}} = 1$$

$$\omega_{gx1} = 2 \text{ MHz}$$

$$PM = \cancel{180} = 180 - \tan^{-1} \frac{\omega_{gx1}}{\omega_{p1}} - \tan^{-1} \frac{\omega_{gx1}}{\omega_{p2}}$$

$$\omega_{gx2} = 4.05 \text{ MHz}$$

PM

- (b) A unity-gain closed-loop amplifier exhibits a frequency peaking of 50% in the vicinity of the gain crossover. What is the phase margin?

Phase margin	33.69°
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Important Steps:

$$\omega = 1.5 \omega_{gx}$$

$$PM = 90 - \tan^{-1} \omega$$